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SIGNAL PROCESSING FOR MEMS SENSOR BASED MOTION ANALYSIS SYSTEM

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Abstract

Sensor systems for motion analysis represent an important class of embedded sensor systems for health, and are usually based on MEMS technology (Micro-electro-mechanical systems). Gyroscopes and accelerometers are two examples of MEMS motion sensors that are characterized by their small size, low weight, low power consumption, and low cost. This makes them suitable to be used in wearable systems, intended to measure body movements and posture, and to provide the input for advanced human motion analyses. However, MEMS sensors usually are sensitive to environmental disturbances, such as shock, vibration and temperature changes. A large portion of the measured MEMS sensor signal actually originates from error sources such as noise, offset, and drift. Especially, temperature drift is a well-known error source. Accumulation errors increase the effect of the error during integration of acceleration or angular rate to determine the position or angle. Thus, methods to limit or eliminate the influence of the sources of errors are urgent. Due to MEMS sensor characteristics and the measurement environment in human motion analysis, signal processing is regarded as an important and necessary part of a MEMS sensor based human motion analysis system.

This licentiate thesis focuses on signal processing for MEMS sensor based human motion analysis systems. Different signal processing algorithms were developed, comprising noise reduction, offset/drift estimation and reduction, position accuracy and system stability. Further, real time performance was achieved, also fulfilling the hardware requirement of limited calculation capacity. High-pass filter, LMS algorithm and Kalman filter were used to reduce offset, drift and especially temperature drift in a MEMS gyroscope based system, while low-pass filter, LMS algorithm, Kalman filter and WFLC algorithms were used for noise reduction. Simple methods such as thresholding with delay and velocity estimation were developed to improve the signal during the position measurements. A combination method of Kalman filter, WFLC al-

gorithm and thresholding with delay was developed with focus on the static stability and position accuracy of the MEMS gyroscope based system. These algorithms have been implemented into a previously developed MEMS sensor based motion analysis system. The computational times of the algorithms were all acceptable. Kalman filtering was found efficient to reduce the problem of temperature drift and the WFLC algorithm was found the most suitable method to reduce human physiological tremor and electrical noise. With the Trapezoidal method and low-pass filter, threshold with delay method and velocity estimation method reduced integrated drift in one minute by about 20 meters for the position measurements with MEMS accelerometer. The threshold with delay method made the signal around zero level to zero without interrupting the continuous movement signal. The combination method of Kalman filter, WFLC algorithm and threshold with delay method showed its superiority on improving the static stability and position accuracy by reducing noise, offset and drift simultaneously, 100% error reduction during the static state, 98.2% position error correction in the case of movements without drift, and 99% with drift.

Sammanfattning

Sensorsystem för rörelseanalys utgör en viktig klass av inbyggda sensorsystem för hälsa, och är vanligtvis baserade på MEMS-teknologi (Mikroelektromekaniska system). Gyroskop och accelerometrar är två exempel på MEMS-rörelsesensorer som kännetecknas av sin ringa storlek, låga vikt, låga energiförbrukning och låga kostnad. Detta gör dem lämpliga att användas i bärbara system, avsedd att mäta kroppsrörelser och kroppsställning, och för att ge ingångsvärden till avancerad rörelseanalys hos människor. MEMS-sensorer är dock ofta känsliga för miljöstörningar, såsom stötar, vibrationer och temperaturförändringar. En stor del av den uppmätta MEMS-sensorsignalen härstammar egentligen från felkällor såsom brus, offset och drift. Speciellt utgör temperaturdrift en vanlig felkälla. Ackumulering av felet sker vid integration av acceleration eller vinkelhastighet för att bestämma positionen eller vinkeln. Detta innebär att metoder för att begränsa eller eliminera påverkan av felkällor är av största vikt. MEMS-sensorernas egenskaper och den mätmiljö som rörelseanalys innebär gör att signalbehandling utgör en både viktig och nödvändig del av ett MEMS-sensorbaserat rörelseanalyssystem.

Denna licentiatavhandling fokuserar på signalbehandling av MEMS-sensorbaserade rörelseanalyssystem. Olika signalbehandlingsalgoritmer har utvecklats för brusreducering, offset- / driftuppskattning och reducering, positioneringssnoggrannhet och systemstabilitet. Vidare har realtidsprestanda uppnåtts som också uppfyller hårdvarukravet på begränsad beräkningskapacitet. Högpasfilter, LMS-algoritm och Kalman-filter har använts för att minska offset, drift och särskilt temperaturdrift i ett MEMS-gyroskopbaserat system, medan lågpasfiltret, LMS-algoritm, Kalmanfilter och WFLC algoritm har använts för brusreducering. Enkla metoder såsom tröskling med fördröjning och hastighetsuppskattning har utvecklats för att förbättra signalen vid positionsmätning. En kombinationsmetod som kombinerar Kalman filter, WFLC algoritm och tröskling med fördröjning har utvecklats för att åstadkomma

statisk stabilitet och lägesnoggrannhet för ett MEMS-gyroskopbaserat system. Dessa algoritmer har implementerats i ett tidigare utvecklat MEMS-sensorbaserat rörelseanalyssystem. Algoritmernas beräkningstider var acceptabla. Kalmanfiltrering var effektivt för att minska problemet med temperaturdrift och WFLC algoritmen utgjorde den mest lämpliga metoden för att minska fysiologisk tremor och elektriska störningar. Med Trapetsmetoden och lågpassfilter, tröskling med fördröjning och hastighetsberäkning minskade den integrerade driften en minut med ca 20 meter för positionsmätning med MEMS-accelerometer. Tröskling med fördröjning lade signalen kring nollnivå till noll utan att avbryta den kontinuerliga signalen. Kombinationsmetoden Kalman filter, WFLC algoritmen och tröskling med fördröjning visade sig mest lämpad för att förbättra statisk stabilitet och positioneringsnoggrannhet genom att minska brus, offset och drift samtidigt, 100 % felminskning under statiskt tillstånd, 98,2 % lägre felkorrigering vid rörelser utan drift, och 99 % med drift.

To my family

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Jiaying Du
Västerås, March 21, 2016

List of Publications

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Paper B *Signal Processing algorithms for position measurement with MEMS-based accelerometer.* Jiaying Du, Christer Gerdtnan and Maria Lindén. In IFMBE Proceedings of 16th Nordic-Baltic Conference on Biomedical Engineering, Vol.48, page(s): 36–39. October, 2014.

Paper C *Noise reduction for a MEMS-gyroscope based head mouse.* Jiaying Du, Christer Gerdtnan and Maria Lindén. In Proceeding of 12th International Conference on Wearable Micro and Nano Technologies for Personalized Health, page(s): 98–104. June, 2015.

Paper D *A signal processing algorithm for improving the performance of a gyroscopic computer head mouse.* Jiaying Du, Christer Gerdtnan, Arash Gharehbaghi and Maria Lindén. Submitted to Journal.

¹The included papers have been reformatted with the licentiate thesis layout.

Related work, Not Included in the Licentiate Thesis:

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- *Signalbehandlingshäftet till TDG-systemet*. Jiaying Du, Christer Gerdman. Motion Control. 2014.
- *Investigation for measurements of body parameters with sensors*. Jiaying Du, Christer Gerdman, Maria Lindén. Micronano System Workshop 2014 (MSW2014). May, 2014.
- *Patentundersökning: KWT-algoritmen*. Jiaying Du, Christer Gerdman. Motion Control. 2015.
- *Signal processing with application to a MEMS-gyroscope based computer head mouse*. Jiaying Du, Christer Gerdman, Maria Lindén. Submitted to Micronano System Workshop 2016 (MSW2016).
- *The effects of perceived USB-delay for sensor and embedded system development*. Jiaying Du, Daniel Kade, Christer Gerdman, Oguzhan Özcan, Maria Lindén. Submitted to 38th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC2016).

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List of Abbreviations

ADC	Analog Digital Converter
ALP	Adaptive Linear Prediction
ANC	Adaptive Noise Cancellation
DAC	Digital Analog Converter
DSP	Digital Signal Processor
dsPIC	Digital Signal Controller
FLC	Fourier Linear Combiner
HPF	High-pass Filter
HCI	Human-Computer Interface
KF	Kalman Filter
KWT	The combination of KF, WFLC and TWD
LMS	Least Mean Square
LPF	Low-pass Filter
MEMS	Micro-electro-mechanical system
PIC	Peripheral Interface Controller
RMS	Root Mean Square
RMSE	Root Mean Square Error
TWD	Threshold with Delay
μP	Microprocessor
USB	Universal Serial Bus
WFLC	Weighted-frequency Fourier Linear Combiner

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I

Thesis

Chapter 1

Introduction

Human motion analysis is an important research topic. Especially it can be applied in the area of healthcare, such as rehabilitation, athletic performance analysis, monitoring of health for the elderly at home etc. [1]. A variety of systems have been developed for human motion analysis. Using cameras is a common way for motion capture [2]. This has been developed with several or single three-dimensional (3D) cameras [3] [4]. However, a free-living measurement environment has been an extremely challenging issue [5]. The wearable health monitoring system has attracted considerable attention from both industry and research community in the recent years, as the increasing costs of healthcare, demographic changes and multiple diseases [6]. This addresses the demand where miniature, low-cost and low-weight are appreciated [7]. Micro-electro-mechanical systems (MEMS) represents a class of miniature electro-mechanical devices. The MEMS technology has rapidly developed in the latest 50 years and has been applied in many areas, such as consumer products, aerospace, automotive, biomedical, chemical, optical displays, wireless and optical communications and fluidics etc., where miniaturization is beneficial [8]. MEMS based motion sensors such as MEMS gyroscopes and MEMS accelerometers are a kind of inertial sensors [9]. They are in micro-scale. Compared to the traditional ones, they usually have the advantage of small size, light weight, low cost, low power consumption, high sensitivity, good linearity and high precision [10]. Therefore, they have been applied extensively in the areas of automobiles, consumer electronics, spacecraft and robotics [11], especially in the consumer-grade application of motion interface [12].

The MEMS gyroscope and MEMS accelerometer have already been widely

used in the motion analysis system in medical field, for example, the knee/ankle joint measurement [13] [14] [15] [16], gait analysis [17] [18], ambulatory measurement and analysis of the lower limb [19] [20], collection of anatomical joint angles during stair ascent [21], hand gesture recognition [22] [23] [24], head motion controlled wheel chair [25], and head motion controlled mouse [26] [27]. They were also adopted to develop a digital motion analysis system for rehabilitation from impairment such as those caused by accidents or stroke [28] [29].

In previous research work, two different motion detection systems have been developed. One was a MEMS gyroscope based computer head mouse (MultiPos system) to help the persons with mobility disorders [27]. The other was a 3D sensor system combining MEMS gyroscope and MEMS accelerometer (TDG system) for motion measurement [30]. The MEMS gyroscope senses the angular rate, while the MEMS accelerometer detects the acceleration. The measured signals can be used for position detection. However, due to the sensitivity of the MEMS sensors to environmental disturbances, such as shock, vibration and temperature change, a large portion of the measured MEMS sensor signals actually origin from errors such as noise, offset and drift. Accumulated drift error is also a common problem during integration process to determine the position or angle.

Signal processing is one of the key technologies that ensure the stability and reliability of the systems. By now many methods have been applied for the sensor compensation. They can be classified into two major kinds, analog signal processing and digital signal processing [31]. Analog filters are used as anti-alias filter and reconstruction filter that placed before Analog-Digital Converter (ADC) and after Digital-Analog Converter (DAC) in the digital signal processing system [32]. Generally digital methods including different signal processing algorithms are cost-efficient to solve the sensor problems. Low-pass filter is a common and simple algorithm for the noise reduction. It can reduce the high-frequency noise [33]. Least Mean Square (LMS) algorithm is a basic adaptive algorithm, by which both adaptive noise cancellation filter and adaptive linear prediction filter can be implemented [34]. The Kalman filter is the most common algorithm for the sensor problem such as gyro sensor drift compensation [35]. It was also applied for human tremor estimation [36] [37]. Extended Kalman filter (EKF) has been developed for improving the accuracy of MEMS inertial sensors [38]. A wavelet based method like wavelet threshold denoising method has been improved and applied to eliminate random noise of MEMS gyroscope [39]. Weighted Fourier Linear Combiner (WFLC) algorithm has been used for adaptive human-machine interface for persons with

tremor [40]. It was also used to estimate displacement of periodic motion with inertial sensors [41]. Adaptive band-pass filter is a basic algorithm to reduce the typical noise, tremor from inertial sensor data [42].

With the requirement of the developed hardware system at hand, more suitable and efficient methods are desired in this study. The research does not only include the development of signal processing algorithms, but also the application of signal processing algorithms to the related sensor problems including in the sensor system are compared and analysed.

1.1 Motivation

MEMS sensor based motion analysis system represents one important class of embedded sensor systems for measuring physiological parameters, in which inertial sensors such as MEMS gyroscope and MEMS accelerometer are used for different body movement measurements and position measurements. Inertial sensors are useful for monitoring of human motion, however, the resolution and stability of MEMS inertial sensors are not adequate and need to be improved [43]. The common errors can be classified into noise, offset and drift. Temperature drift, a kind of drift caused by temperature changes, is a typical error in the MEMS gyroscope based system. The accumulated error during integration process is also a critical problem in the position measurements. The features of miniature and low-cost made the wearable sensor system consist of a single microprocessor or microcontroller, programmable logic device or application-specific integrated circuit, and also need to perform real-time signal processing [44]. It is not a trivial task considering the limitations in arithmetic power, memory and power consumption resources [44]. To fulfil the hardware requirements of limited calculation capacity in real time, there is a need to ensure that the sensor data can be used correctly. Depending on the hardware of the developed system at hand, more suitable algorithms are required to improve the signal quality and the system performance.

1.2 Research hypothesis and questions

The signals from MEMS gyroscopes and MEMS accelerometers usually consist of a large portion of noise, offset and drift, especially temperature drift. Further, to get a position or angle from the measured MEMS accelerometer or MEMS gyroscope signals, it is of importance to handle the error signal that accumulate over time. The conversion to the position can give rise to large

measurement errors. In addition, the developed embedded sensor systems have limited calculation capacity in real time. Thus, the hypothesis is:

- *It is possible to implement suitable signal processing algorithms to the corresponding developed sensor system with the requirements of system and limitation of hardware so that the MEMS gyroscope/MEMS accelerometer signals are improved and accurate enough for further analyses.*

From the research hypothesis, two concrete research questions are formulated in this thesis, which are mostly about the research area of signal processing for embedded sensor systems with MEMS gyroscopes and MEMS accelerometers.

- **Q1:** *How can the quality of MEMS gyroscope/MEMS accelerometer signals be improved, including minimizing the effect of noise and temperature drift?*

The MEMS gyroscope and MEMS accelerometer can be used to measure the movement of any body part in order to get body movement parameter. They can be applied in the area of rehabilitation treatment or help models. The noise is the most common problem in signal processing. The noise in the MEMS gyroscope or the MEMS accelerometer signals has its own characteristics. For example, the human tremor can be seen as a typical noise during the movement measurement with MEMS gyroscopes and MEMS accelerometers. Temperature drift is a typical problem for MEMS gyroscope/MEMS accelerometer signals. How to reduce most of the noise and compensate the temperature drift are investigated in this thesis.

- **Q2:** *How can an accurate position measurement be accomplished by MEMS gyroscope and MEMS accelerometer?*

The MEMS accelerometer measures acceleration while the MEMS gyroscope measures angular rate. The velocity, position and angle can be derived by integration. However, during the calculation and conversion from the acceleration/angular rate to the velocity, position/angle, there are errors and deviations. So how the position measurements by MEMS gyroscope and MEMS accelerometer are processed in order to be accurate enough is also a concerned research question.

1.3 Outline of the thesis

This thesis is organized as two main parts, i.e. thesis and included papers. The thesis part is divided into six chapters. The first chapter presents motivation of the research work, research hypothesis and questions. The second chapter introduces the background of the research. Chapter 3 includes the descriptions of signal processing algorithms in this research work. The fourth chapter describes the scientific contributions of each included papers. Chapter 5 presents the result and discussion. The last chapter concludes the dissertation and gives an insight into the possible future work. In the included papers part, four papers are presented and organized in the form of chapters from 7 to 10.

Chapter 2

Background

The research work was performed with the collaboration between the company Motion Control i Västerås AB, the research profile Embedded Sensor System for Health (ESS-H) and the ITS-EASY Post Graduate School for Embedded Software and Systems (ITS-EASY) at MDH.

Motion Control develops wearable products with sensors. The typical examples of the systems are the gyroscope based computer head mouse for disabled persons, the MultiPos system, as shown in [27], and a 3D sensor measurement system that called TDG sensor system [30]. This thesis focuses on the development of signal processing algorithms to achieve better performance of these two systems, and also have the potential to be used in similar systems.

2.1 MEMS sensor based motion analysis systems

2.1.1 MultiPos system

The MEMS sensor based computer head mouse, i.e. the MultiPos system, is an embedded system with MEMS gyroscopes that can detect head movement. It has been developed aiming to help persons with movement disorders [27]. It enables the users to control the computer cursor by moving the head instead of the hands. As shown in Figure 2.1, there are three main parts in the MultiPos system. The sensor part is a wearable head band with MEMS gyroscope. The central part includes communication and control functions. The part for click function can be selected among click device, sip-and-puff device and biting device depending on different user needs.



Figure 2.1: MultiPos system.

In order to catch the head movement, the MEMS gyroscope is used to measure rotation in x-axis and y-axis. However, because of the sensitivity of MEMS gyroscope to the environmental disturbances such as shocks, vibrations and temperature changes, a large portion of noise such as other movements like vibration and tremor, environment factors like temperature changes etc., are also included in and have big effect on the MEMS gyroscope signal.

The MultiPos system works in real time with limited calculation capability due to the hardware limitation of the MultiPos system. There is a dsPIC in the MultiPos system used as a slave processor with signal processing functions. Signal processing algorithms should be designed to improve the MultiPos signal quality with respect to noise influence.

2.1.2 TDG sensor system

The 3D motion measurement system named the TDG sensor system is an embedded sensor system for measuring movements. The intended application are healthcare, human motion analysis and rehabilitation support. The TDG system evaluates a person's ability to move in different body parts such as neck, arm or leg. It is aimed to training or rehabilitation, e.g. for patients who have suffered from whiplash injuries due to traffic accidents. The system was developed with different MEMS gyroscopes and MEMS accelerometers that are

fixed in the different forms of wearable bands around head, arm or leg. As shown in Figure 2.2, the system consists of different sensors and a central part for communication and control. It is connected to a computer via the USB-port.



Figure 2.2: TDG system.

The sensor part can be selected among 1D, 2D and 3D MEMS gyroscopes and MEMS accelerometers. The MEMS gyroscope and MEMS accelerometer measure the movement to get the position and movement pattern for motion analysis. MEMS gyroscope measures angular rate, from which the angular position can be derived by integration, while MEMS accelerometer measures acceleration so that the velocity and position can also be calculated. However, the errors in the measured signals such as noise, offset and drift were enlarged by digital integration. Except for the errors from sensor itself, the integration drift is a problem that need to be solved. Thus, the signal processing is desired to improve the signal quality and accuracy of position or angle calculation.

2.2 MEMS sensor signals

Both MEMS gyroscope and MEMS accelerometer are used in the developed motion analysis system. There are three axes for 3D MEMS gyroscope and MEMS accelerometer signals, i.e. x , y and z . The z -axis basement of 3D MEMS gyroscope and MEMS accelerometer signals are different because of their working principles.

MEMS gyroscope

The MEMS gyroscope is based on the Coriolis effect to measure the angular rate [45]. The working principle of a 3D MEMS gyroscope is shown in Figure 2.3. There are three-axis signals x, y and z. All the three-axis signals are zero in the ideal case when it is stationary. When it is moved, the MEMS gyroscope detects the angular rate and the corresponding signal is in the unit of degree per second [$^{\circ}/s$].

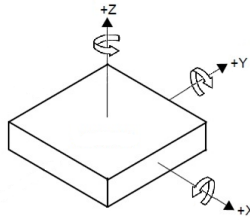


Figure 2.3: 3D MEMS gyroscope.

An example of three-axis MEMS gyroscope signal is shown in Figure 2.4. The blue, red and green curve present x-, y- and z-axis signal, respectively. Movement is between 3 s and 20 s. Without large drift and offset, the stationary x-, y- and z-axis signals are all around zero.

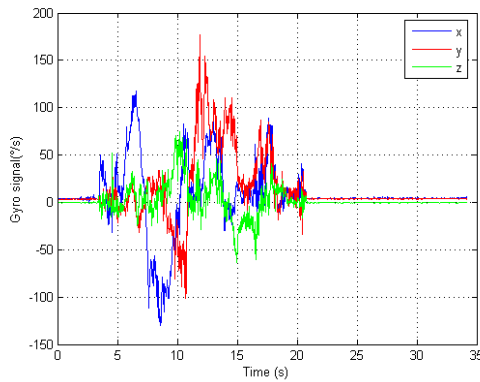


Figure 2.4: 3D MEMS gyroscope signal.

MEMS accelerometer

The MEMS accelerometer measures acceleration. As shown in Figure 2.5, earth gravity which is at the opposite direction of z-axis affects the z-axis signal of the MEMS accelerometer. Thus, the z-axis signal of the MEMS accelerometer is $1g$ (where $g = 9.81m/s^2$); while the x- and y-axis signals are zero in the ideal case without movement. The movement MEMS accelerometer signal is acceleration signal in $[m/s^2]$.

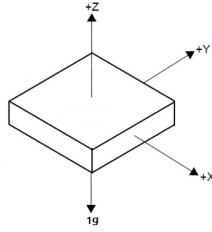


Figure 2.5: 3D MEMS accelerometer.

An example of three-axis MEMS accelerometer signal is shown in Figure 2.6. The blue, red and green curve present x-, y- and z-axis signal, respectively. Movement is from 3 s to 20 s. Without large drift and offset, the x- and y-axis signals are around zero and z-axis signal is around 1 g.

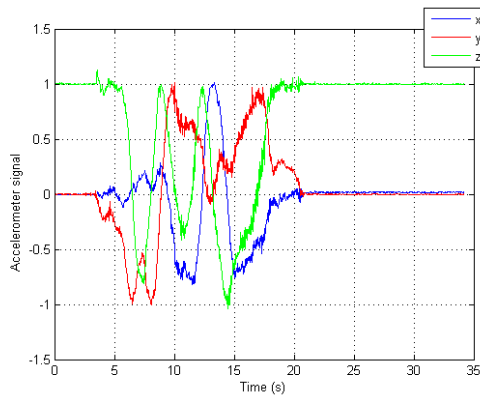


Figure 2.6: 3D MEMS accelerometer signal.

2.3 MEMS sensor signal errors

The sensor error can be noise, offset, drift, hysteresis, scale factor error, non-linearity and accuracy error. In this study, the signal problems of the motion sensor system are classified as three main categories, which are noise, offset, drift as shown in Figure 2.7.

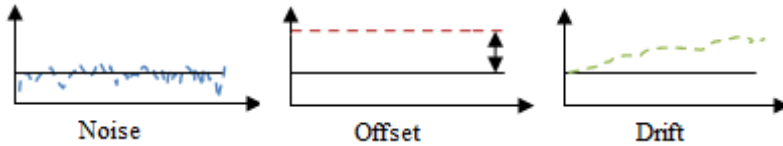


Figure 2.7: Noise, offset and drift.

Noise presents in almost all the measured sensor signals. The existing noise sources in the developed motion sensor systems are summarized as:

- Noise from the environment like temperature fluctuations, gravity, etc.
- Unwanted movements, for example, shocks, vibrations and human tremor which include both physiological tremor and pathological tremor.
- Noise from the hardware such as electrical noise and mechanical thermal noise.

Offset error, is also called zero error or bias error. It is characterized that the output of the sensor is constant when there is no measured value. This implies a shift or displacement of the zero point. It is a common error that exists with most of the measured sensor signals.

Drift means that the output changes with time when there is no change of the input signal. Drift can be caused by vibration, shock, aging, and temperature variations. Temperature drift is generated due to the temperature changes. This phenomenon is well-known with the MEMS gyroscope [46]. Accumulation error can also be regarded as a drift due to numerical integration during the position calculation [47].

Due to the MEMS sensor signal errors in the embedded sensor systems, the signal processing in this study was focusing on the noise reduction, offset/dead zone and drift estimation and reduction, position accuracy and signal stability.

Chapter 3

Signal processing algorithms

Different signal processing algorithms were developed and applied to improve the signal quality of the MultiPos system with the related problem. Low-pass filter [48] [49], high-pass filter [48], LMS algorithm [34], Kalman filter [35], WFLC algorithm [40] were applied into the systems to solve the MEMS sensor problems with noise, offset, drift and temperature drift. Threshold with delay (TWD) method and velocity trend estimation method were designed to solve the problems during the position measurements with MEMS sensors. TWD method were also used to improve the static stability. The signal processing algorithms were simplified and optimized due to the hardware limitation. With different MEMS sensors and different signal errors, these simplified filter functions can be used either by itself or in a combination with others to reduce the signal errors and improve the signal quality. The details of the signal processing algorithms are introduced in this chapter.

3.1 Low-pass filter

A simple low-pass filter can be a digital implemented RC low-pass filter [49]. The low-pass filter has its cut-off frequency adapted to let the appropriate frequencies below pass. The cut-off frequency can be described as a parameter with a value from zero to one. This kind of low-pass filter was applied into the MEMS gyroscope and MEMS accelerometer based motion detection system for noise reduction.

The low-pass filter can be described as [49]:

$$y_i = y_{i-1} + \alpha(x_i - y_{i-1}) \quad (3.1)$$

Where α is the parameter of cut-off frequency for the low-pass filter. x_i is input signals at time point i . y_i and y_{i-1} are output signals at time point i and $i - 1$, respectively.

3.2 High-pass filter

In analogy to the digital implementation of a RC low-pass filter [49], high-pass filter can be implemented in a similar way. The high-pass filter has its cut-off frequency adapted to let the appropriate frequencies above pass. Here the cut-off frequency is also described as a parameter with a value from zero to one.

The high-pass filter can be described as [50]:

$$y_i = \beta y_{i-1} + \beta(x_i - x_{i-1}) \quad (3.2)$$

Where β is the parameter of cut-off frequency for the high-pass filter. x_i and x_{i-1} are input signals at time point i and $i - 1$ respectively. y_i and y_{i-1} are output signals at time point i and $i - 1$ respectively.

3.3 LMS algorithm

Adaptive filters are required for certain applications where some parameters of the desired signal processing are not known in advance. The adaptive filter uses feedback in the form of an error signal to refine its transfer function to match the changing parameters. The adaptive filter using LMS algorithm can be described as follows [34].

Input signal:

$$x(n) = [x(0), x(1), \dots, x(n-1)]^T \quad (3.3)$$

Adaptive filter coefficient vector:

$$w_n = [w(0), w(1), \dots, w(n-1)]^T \quad (3.4)$$

$$\Delta w_n = \mu e(n) x^*(n) \quad (3.5)$$

μ is the step size and converges to

$$0 < \mu < \frac{2}{\lambda_{max}} \quad (3.6)$$

Where λ_{max} is the largest eigenvalue to the regression covariance matrix.

LMS-based adaptive filter has $P + 1$ coefficients and the updating adaptive filter is

$$w_{n+1} = w_n + \mu \Delta w_n = w_n + \mu e(n) x^*(n) \quad (3.7)$$

And the output of the adaptive filter is

$$y(n) = \sum_{i=0}^P w_n(i) x(n-i) \quad (3.8)$$

Then it gives

$$e(n) = d(n) - y(n) = d(n) - W(n) x(n) \quad (3.9)$$

LMS algorithm can be used to reduce noise or drift. It is an adaptive noise canceller (ANC) using LMS algorithm for the reduction of noise, while an adaptive linear prediction filter (ALP) using the LMS algorithm for the reduction of drift and offset. The structure of the adaptive filter is shown in Figure 3.1, where $x(n)$ is the input signal, $d(n)$ is the reference signal, $y(n)$ is the output signal of adaptive filter, $e(n)$ is the error signal, and W is the adaptive filter with LMS algorithm.

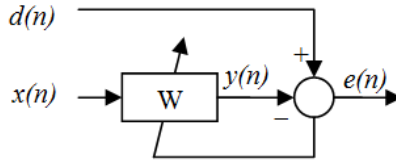


Figure 3.1: Adaptive filter with LMS algorithm.

3.4 Kalman filter

A Kalman filter is a recursive filter used to estimate the state of a process. Some common applications of Kalman filter are reduction of noise, signal prediction

and signal estimation. There are two main processes in the Kalman filter algorithm. One is time updating, which is a process of prediction and the other is measurement updating, which is process of correction.

Time updating for discrete Kalman algorithm [51]:

$$\hat{x}_k^- = A\hat{x}_{k-1} + Bu_{k-1} \quad (3.10)$$

$$P_k^- = AP_{k-1}A^T + Q \quad (3.11)$$

Measurement updating for discrete Kalman algorithm [51]:

$$K_k = P_k^- H^T (HP_k^- H^T + R)^{-1} = \frac{P_k^- H^T}{HP_k^- H^T + R} \quad (3.12)$$

$$\hat{x}_k = \hat{x}_k^- + K_k (z_k - H\hat{x}_k^-) \quad (3.13)$$

$$P_k = (I - K_k H) P_k^- \quad (3.14)$$

Where A is state transition model, B is control-input model, H is measurement matrix, I is identity matrix, K is Kalman gain, k is discrete time point, P is error covariance, Q is process noise covariance, R is measure noise covariance, u is control vector, x is the state, $\hat{x}$ is the posteriori estimated state, $\hat{x}_k^-$ is priori state, and z is the measurement. The process can also be seen in Figure 3.2 [51].

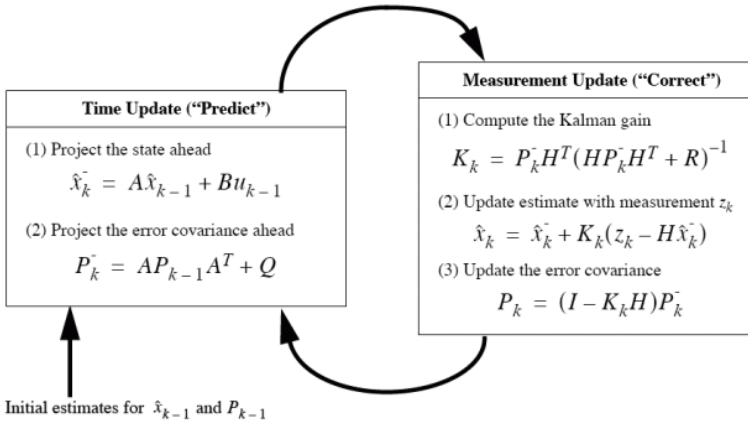


Figure 3.2: Kalman filtering.

A Kalman filter can be used for noise reduction. It can also be used to estimate the drift or offset trend so that the drift and offset can be reduced. Structure of the application of Kalman filter for the reduction of drift and offset is shown in Figure 3.3, where $x(n)$ is the input signal, $u(n)$ is the output signal of the Kalman filter, $y(n)$ is the output signal after drift and offset reductions, and KF is Kalman filter.

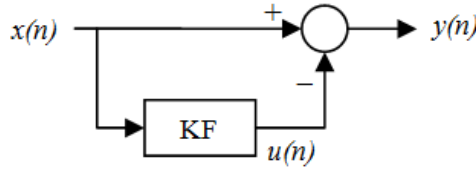


Figure 3.3: Application of a Kalman filter for drift/offset reduction.

3.5 WFLC algorithm

Physical tremor is natural in all human beings. It exists in all human with amplitude lying in the frequency range of 8–12 Hz [52].

WFLC is an adaptive signal processing algorithm that models a truncated Fourier series model of a period signal with unknown frequency and amplitude. The tremor noise is modulated as a sinusoid and the modulation is tracked in frequency, phase and amplitude. The WFLC algorithm is described as follows [40]:

$$x_{rk} = \begin{cases} \sin \left(r \sum_{t=0}^k w_0 \right) & 1 \leq r \leq M \\ \cos \left((r - M) \sum_{t=0}^k w_0 \right) & M + 1 \leq r \leq 2M \end{cases} \quad (3.15)$$

$$y_{k+1} = y_k + W^T X \quad (3.16)$$

$$\epsilon_k = s_k - y_k - w_{bias_k} \quad (3.17)$$

$$w_{bias_{k+1}} = w_{bias_k} + 2\mu_b \epsilon_k \quad (3.18)$$

$$w_{0_{k+1}} = w_{0_k} + 2\mu_0 \epsilon_k \sum_{i=1}^M (w_i x_{M+i} - w_{M+i} x_i) \quad (3.19)$$

$$W_{k+1} = W_k + 2\mu_1 X_k \epsilon_k \quad (3.20)$$

Where $W_k = [w_{1k}, \dots, w_{2Mk}]^T$, $X_k = [x_{1k}, \dots, x_{2Mk}]^T$. M is the harmonic number. μ_0, μ_1, μ_b are the adaptive parameters. s_k is input signal and ϵ_k is error at time point k .

It can also be described in Figure 3.4 [53].

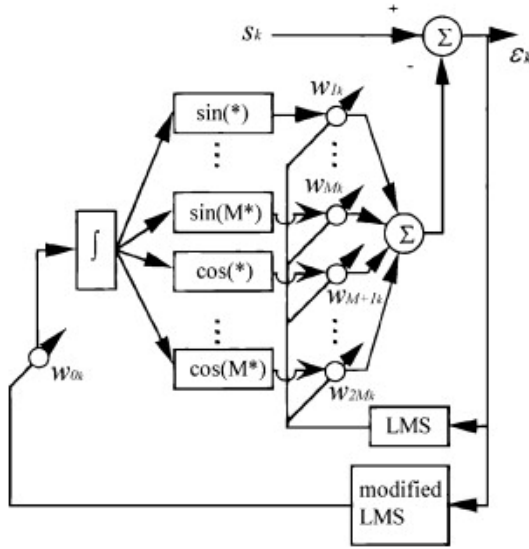


Figure 3.4: WFLC algorithm.

3.6 TWD method

Threshold with delay method, i.e. TWD method, is based on the simple threshold method in which the input signal is set to zero if the value is smaller than threshold. However, the drawback of the threshold method is that there is often not a smooth transition at the threshold setting point and the signal becomes choppy. Thus, the TWD method is designed to improve the signal under threshold such as continuous movement signal. The comparison of normal threshold method and TWD method is shown in Figure 3.5 without noise and Figure 3.6 with noise.

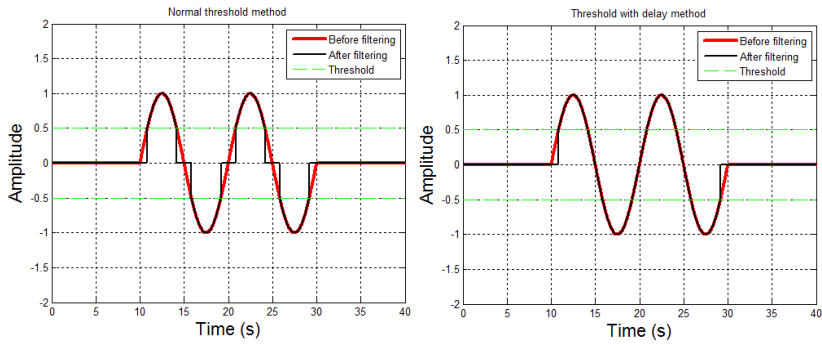


Figure 3.5: Comparison of normal threshold method and TWD method without noise ($T=0.5$).

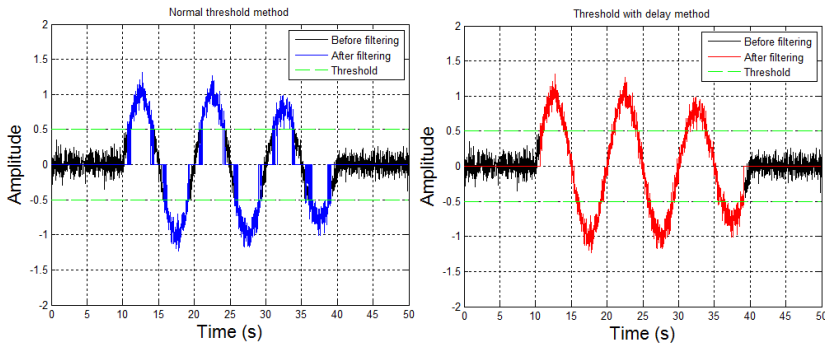


Figure 3.6: Comparison of normal threshold method and TWD method with noise ($T=0.5$).

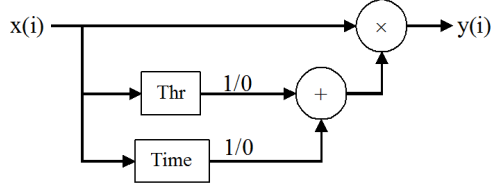


Figure 3.7: TWD method.

As shown in Figure 3.7, the algorithm can be described as:

$$y_i = \begin{cases} x_i, & x_i \geq T \text{ or } t_{x \leq T} < D \\ 0, & x_i < T \text{ and } t_{x \leq T} > D \end{cases} \quad (3.21)$$

Where T is the parameter of threshold and D is the parameter of delay. x_i is input signal at time point i . y_i is output signal at time point i . $t_{x \leq T}$ is the accumulated time while input signal is small than threshold value.

3.7 Velocity trend estimation

Velocity drift estimation method is designed according to the features of velocity signal which is similar to a sinus signal.

The velocity drift estimation method can be described as:

$$y_i = \begin{cases} x_i, & A_i = 0 \\ y_{i-1}, & A_i \sim 0 \end{cases} \quad (3.22)$$

Where x_i is input signal at time point i , and y_{i-1} is output signal at time point $i - 1$. A_i is derivation of input signal which can be obtained from:

$$A_i = K \times \text{abs}(x_i - x_{i-1}) \quad (3.23)$$

Where K is a scale parameter.

3.8 KWT algorithm

The KWT algorithm is a combination algorithm of Kalman, WFLC and TWD. It takes the advantage of Kalman filter, WFLC algorithm and TWD method.

It is developed with the aim to reduce noise, offset and drift simultaneously. The Kalman filter is used for drift/offset estimation. The WFLC is for noise reduction. The TWD method is to provide a good static zero-level signal before and after the movement. The measured signal minus estimated drift/offset by Kalman filter and then denoised by WFLC and TWD. The main structure of the algorithm is shown in Figure 3.8.

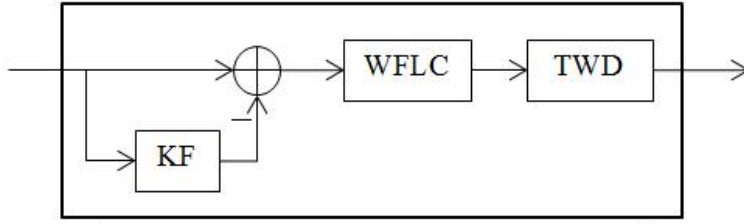


Figure 3.8: Structure of KWT algorithm.

3.9 Position measurement

MEMS gyroscopes and MEMS accelerometers can be used to sense the movement by measuring angular rate [$^{\circ}/s$] and acceleration [m/s^2] respectively.

For the accelerometer signal, the velocity can be calculated by integrating the acceleration and then the velocity is integrated to get the position in meters [m].

$$s = \int \vec{v} dt = \int \left(\int \vec{a} dt \right) dt \quad (3.24)$$

For the MEMS gyroscope signal, the angular position [$^{\circ}$] can be got by integration of the measured angular rate signal.

$$\theta = \int \vec{\theta} dt \quad (3.25)$$

Because of the discrete digital data and error in the integration of digital data, the Trapezoidal method [54] is used to get a more accurate result of integration.

For MEMS accelerometer, the integration calculation is:

$$v_i = v_{i-1} + \left(a_{i-1} + \frac{(a_i - a_{i-1})}{2} \right) \times T \quad (3.26)$$

$$s_i = s_{i-1} + \left(v_{i-1} + \frac{(v_i - v_{i-1})}{2} \right) \times T \quad (3.27)$$

For MEMS gyroscope, the integration calculation is:

$$\theta_i = \theta_{i-1} + \left(\dot{\theta}_{i-1} + \frac{(\dot{\theta}_i - \dot{\theta}_{i-1})}{2} \right) \times T \quad (3.28)$$

Where a_i is acceleration signal, v_i is velocity signal, s_i is position signal, $\dot{\theta}_i$ is angular rate signal, θ_i is angular position signal at time point i , and a_{i-1} is acceleration signal, v_{i-1} is velocity signal, s_{i-1} is position signal, $\dot{\theta}_{i-1}$ is angular rate signal, θ_{i-1} is angular position signal at time point $i - 1$.

Chapter 4

Scientific contributions

Different signal processing algorithms and methods were investigated, simulated, implemented in the microprocessor and evaluated. Low-pass filter was applied to reduce the high-frequency noise. High-pass filter was introduced to reduce low-frequency interferences such as drift and offset. Least Mean Square (LMS) algorithm and Kalman filter can both be used for noise and drift/offset reductions. Weighted Fourier Linear Combiner (WFLC) is an algorithm developed with focus on the human tremor in the human-computer-interface (HCI). It was applied into the MultiPos system for noise reduction. Comparing to high-pass filter and LMS adaptive filter, Kalman filter was demonstrated as the best method for the reduction of the temperature drift in Paper A. As shown in Paper C, four algorithms, Low-pass filter, LMS algorithm, Kalman filter and WFLC algorithm, were implemented to reduce the noise in the MultiPos system. The WFLC algorithm was demonstrated as the most suitable method to reduce the electrical noise and tremor for the MultiPos system.

Two more methods, threshold with delay (TWD) method and velocity trend estimation method, were designed to solve the problems during the position measurements with MEMS accelerometers in Paper B. About 20 meter integration error was reduced in one minute. A combination algorithm that introduced in Paper D improved the performance of the MultiPos system with respect to noise, offset and drift reduction simultaneously. 100% error reduction during the static state, 98.2% position error correction in the case of movements without drift, and 99% with drift were achieved.

With focus on the two research questions, this thesis presents signal processing algorithms for MEMS sensor based motion analysis systems in the

form of included four papers. The detailed contributions were shown in the following sections of Paper A, B, C and D.

4.1 Paper A

Signal processing algorithms for temperature drift in a MEMS-gyro-based head mouse. *In Proceedings of 21st International Conference on Systems, Signals and Image Processing (IWSSIP2014)*, page(s): 123–126. May, 2014. Authors: Jiaying Du, Christer Gerdman, Maria Lindén.

Abstract:

This paper presents a comparison between different signal processing algorithms applied to a gyro-based computer head mouse for persons with movement disorders. MEMS gyroscopes can be used to sense the head movement and rotation. However, the measured gyro signals are influenced by noise, offset, drift and especially temperature drift. Thus, there is a need to improve the signal by signal processing algorithms. Different gyros have different characteristics and the algorithms should be useful for any selected MEMS gyroscope. In this paper, three different signal processing algorithms were designed and evaluated by simulation in MATLAB and implementation in a dsPIC, with the aim to compensate for the temperature drift problem. The algorithms are high-pass filtering, Kalman algorithm and Least Mean Square (LMS) algorithm. Comparisons and system test show that these filters can be used for temperature drift compensation and the Kalman filter showed the best in the application of a MEMS gyroscope based computer head mouse.

Contribution:

This paper presents signal processing algorithms for the reduction of the temperature drift in the MultiPos system, a MEMS gyroscope based head mouse. It addresses the first research question (Q1) on how to improve the quality of MEMS gyroscope signal by reducing the temperature drift. Several signal processing algorithms and filters such as high-pass filter, LMS algorithm, and Kalman filter were implemented, tested and evaluated. They all can be used to reduce the temperature drift in MultiPos. Kalman filter showed the best performance.

My contribution:

I was the main author of the paper. I did the background study, investigated, simulated, implemented and tested the signal processing algorithms in the paper, analysed the results, and wrote the manuscript.

4.2 Paper B

Signal processing algorithms for position measurement with MEMS-based accelerometer. *In IFMBE Proceedings of 16th Nordic-Baltic Conference on Biomedical Engineering (NBC16), Vol.48, page(s): 36–39. October, 2014.* Authors: Jiaying Du, Christer Gerdtnan, Maria Lindén.

Abstract:

This paper presents signal processing algorithms for position measurements with MEMS accelerometers in a motion analysis system. The motion analysis system is intended to analyze the human motion with MEMS-based sensors which is a part of embedded sensor systems for health. MEMS accelerometers can be used to measure acceleration and theoretically the velocity and position can be derived from the integration of acceleration. However, there normally is drift in the measured acceleration, which is enlarged under integration. In this paper, the signal processing algorithms are used to minimize the drift during integration by MEMS-based accelerometer. The simulation results show that the proposed algorithms improved the results a lot. The algorithm reduced the drift in one minute by about 20 meters in the simulation. It can be seen as a reference of signal processing for the motion analysis system with MEMS-based accelerometer in the future work.

Contribution:

This paper presents the signal processing algorithms/methods for the reduction of accumulated drift of MEMS accelerometer during the position measurement, which addressed the first and second research question (Q1 & Q2) on how to minimize the error and get an accurate position-measurements by MEMS accelerometer. The threshold with delay method and the velocity drift estimation method were developed to reduce the errors from the integration process for position measurement.

My contribution:

I was the main author of the paper. I did the background study, developed the velocity drift estimation method and the flow process, simulated and implemented the methods/algorithms that include low-pass filtering, the threshold with delay method and the velocity drift estimation method, analysed the results, and wrote the manuscript.

4.3 Paper C

Noise reduction for a MEMS-gyroscope-based head mouse. *In Proceeding of 12th International Conference on Wearable Micro and Nano Technologies for Personalized Health (pHealth2015)*, page(s): 98–104. June, 2015. Authors: Jiaying Du, Christer Gerdman, Maria Lindén.

Abstract:

In this paper four different signal processing algorithms which can be applied to reduce the noise from a gyroscope-based computer head mouse are presented. MEMS gyroscopes are small, light, cheap and widely used in many electrical products. MultiPos, a MEMS gyroscope-based computer head mouse system was designed for persons with movement disorders. Noise is a common problem for the MEMS gyroscopes and the performance of MultiPos. In this study four different signal processing algorithms were designed and evaluated by simulation in MATLAB and implementation in a dsPIC, with aim to minimize the noise in MultiPos. The algorithms were low-pass filter (LPF), Least Mean Square (LMS) algorithm, Kalman filter and Weighted Fourier Linear Combiner (WFLC) algorithm. Comparisons and system tests show that these signal processing algorithms can be used to improve the MultiPos system. The WFLC algorithm was found the best method for noise reduction in the application of a MEMS gyroscope based head mouse.

Contribution:

This paper addresses the first research question (Q1) on how to improve the MEMS gyroscope signal by reducing the noise. Four signal processing algorithms, i.e. low-pass filter, LMS algorithm, Kalman filter and WFLC algorithm, were presented in this paper. They were implemented, tested and evaluated regarding the noise problem in MultiPos. The paper demonstrated that

WFLC algorithm was the most suitable for the noise reduction of MultiPos system, especially for the human physiological tremor and electrical noise.

My contribution:

I was the main author of the paper. I did the background study, investigated, simulated, implemented and tested the four signal processing algorithms, analysed the results, and wrote the manuscript.

4.4 Paper D

A signal processing algorithm for improving the performance of a gyroscopic computer head mouse. *Submitted to Journal*. Authors: Jiaying Du, Christer Gerdtnan, Arash Gharehbaghi and Maria Lindén.

Abstract:

This paper presents a signal processing algorithm for removing different noise from a gyroscopic computer head mouse. The proposed algorithm is a combination of Kalman filter (KF), Weighted-frequency Fourier Combiner (WFLC) and threshold with delay method (TWD). The MEMS-gyroscope based computer head mouse has earlier been developed to assist persons with movement disorders. However, since MEMS-gyroscopes usually are sensitive to environmental disturbances as shock, vibration and temperature change, a large portion of noise is added at the same time as the head movement is sensed by the MEMS-gyroscope. The combination method is applied to the MEMS-gyroscopic head mouse, to filter out different types of noise together with the offset and drift, with marginal need of the calculation capacity. The method is examined with both static state test and movement operation test. Angular position is used to evaluate the errors. The results demonstrate that the combination method improved the head motion signal substantially, 100.0% error reduction during the static state, 98.2% position error correction in the case of movements without drift and 99.9% with drift. The proposed combination method in this paper improved the static stability and position accuracy of the MEMS-gyroscopic computer head mouse system by reducing noise, offset and drift, and also has the potential to be used in other gyroscopic sensor systems for signal improvement.

Contribution:

With focus on both the first and second research question (Q1 and Q2) on how to minimize the MEMS gyroscope error and get an accurate angular position, this paper presents a combination method that can improve the MEMS gyroscope signals significantly by reducing noise, offset and drift simultaneously. The combined method was developed with Kalman filter, WFLC algorithm and TWD method. Angular position was used for the evaluation. It gave the MultiPos system more static stability and position accuracy by reducing noise, offset and drift simultaneously. The method of combination different signal processing algorithms also has the potential to be generalized and used in other gyroscopic sensor systems for signal improvement.

My contribution:

I was the main author of the paper. I did the background study, investigated the consisted algorithms, developed, simulated, implemented and evaluated the combination method, analysed the results, and wrote majority of the paper.

Chapter 5

Result and Discussion

Aiming to reduce different signal errors, the signal processing algorithms presented in Chapter 3 were developed and applied in three main aspects, i.e. drift/offset reduction, noise reduction and improvement of position accuracy. The algorithms were simulated in MATLAB® and implemented in a dsPIC®. The performances of algorithms were evaluated in the system test. The measured sensor modules were MEMS gyroscopes IDG300 [55], MLX90609 [56], L3G4200D [57] and MEMS accelerometer ADXL345 [58].

5.1 Signal error reduction

This section addressed the first research question (Q1) on how to minimize signal errors, i.e. drift/offset and noise.

5.1.1 Drift/offset reduction

High-pass filter, LMS algorithm, Kalman filter, and KWT algorithm were adopted for the drift compensation. The results were presented on two sides. One was stationary signal with drift and the other was movement signal with drift. The obvious drift was obtained by large temperature changes from about -5 °C to 21 °C.

Figure 5.1 shows the results of high-pass filtering, LMS filtering, Kalman filtering and KWT filtering of stationary drift signal. Figure 5.2 shows the results of high-pass filtering, LMS filtering, Kalman filtering and KWT filtering of drift signal with movement.

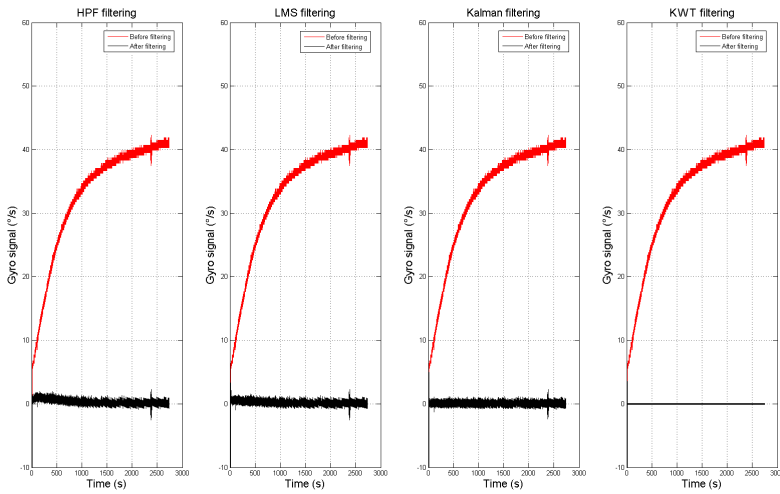


Figure 5.1: Drift reduction without movement by high-pass filter, LMS algorithm, Kalman filter and KWT algorithm respectively.

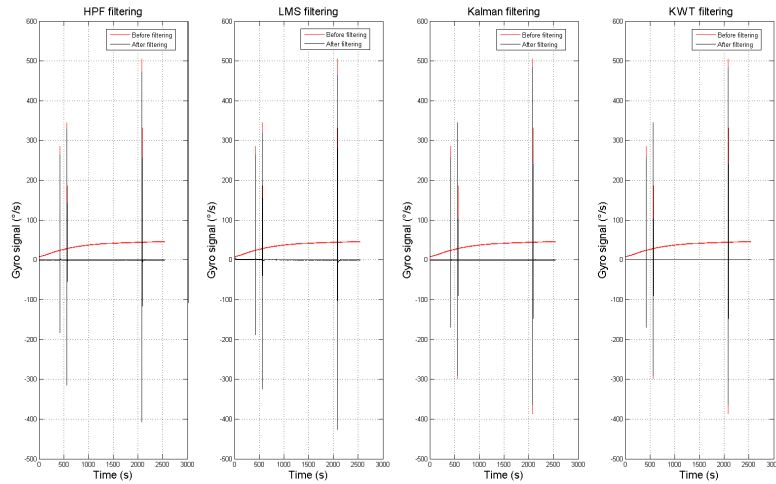


Figure 5.2: Drift reduction with movement by high-pass filter, LMS algorithm, Kalman filter and KWT algorithm respectively.

As shown in Figure 5.1 and Figure 5.2, the signals without filtering are shown as red curves and black curves are the filtered signals. The drift was reduced by all the four algorithms and the movement signals were not eliminated. The KWT algorithm also reduced noise during the static states.

The movement signals were zoomed out. As shown in Figure 5.3, the movement was correctly detected while the drift was reduced.

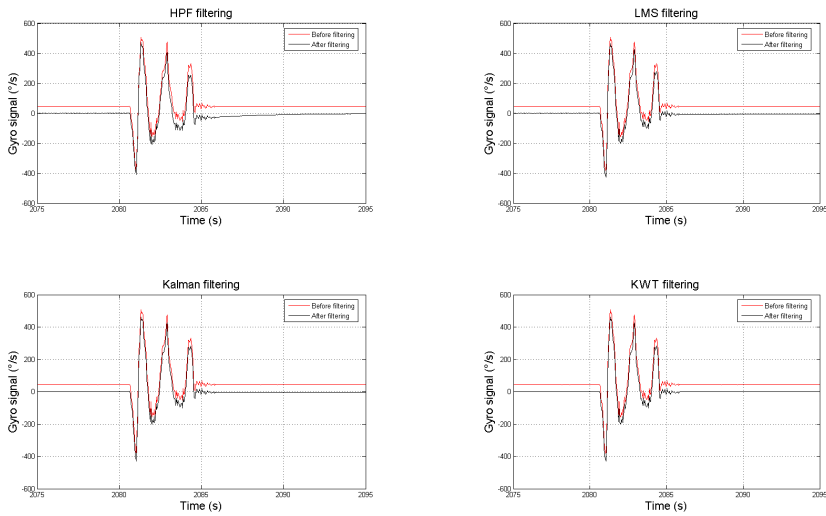


Figure 5.3: Movement during the drift reduction by high-pass filter, LMS algorithm, Kalman filter and KWT algorithm respectively.

As shown in Figure 5.1, Figure 5.2 and Figure 5.3, the four algorithms, i.e. high-pass filter, LMS algorithm, Kalman filter and KWT algorithm can reduce the drift and has neglectable influence on the movement signals. High-pass filter can be easily implemented in a dsPIC. By selection of a suitable cut-off frequency, it can reduce drift and offset effectively. However, there is a slow adjustment at start-up and after the movement. The LMS algorithm and Kalman filter have similar performance for the reduction of drift/offset. The LMS algorithm needs to slightly adjust at start-up. As concluded in Paper A, the Kalman filter was demonstrated as the best algorithm for drift reduction. The KWT algorithm showed its feasibility to reduce drift and offset. It also showed a distinct noise reduction during the static state.

5.1.2 Noise reduction

Low-pass filter, LMS algorithm, Kalman filter, WFLC algorithm and KWT algorithm were adopted to reduce noise. A measured MEMS gyroscope signal without drift/offset was used to evaluate the algorithms. Figure 5.4 presents the noise reduction by low-pass filter, LMS algorithm, Kalman filter, WFLC algorithm and KWT algorithm respectively. Figure 5.5 shows the comparison results of the first 14 seconds stationary signal.

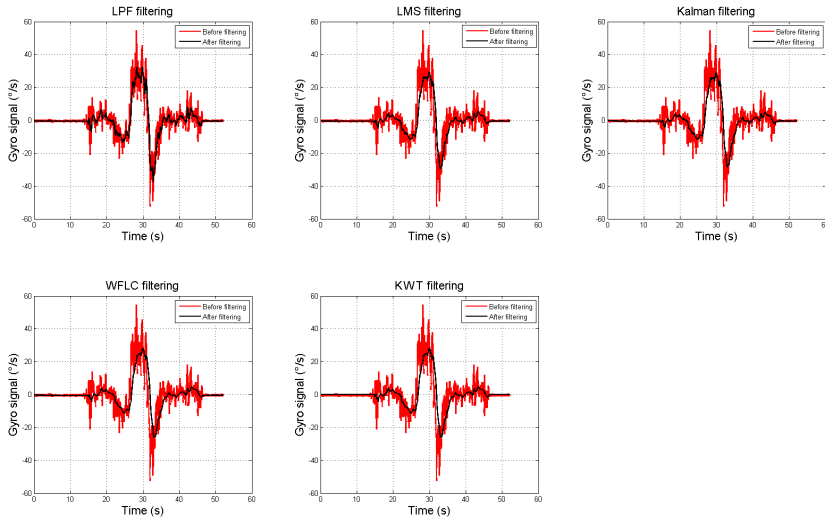


Figure 5.4: Noise reduction with movement by low-pass filter, LMS algorithm, Kalman filter, WFLC algorithm and KWT algorithm, respectively.

The low-pass filter can be easily implemented in the microprocessor. The result of noise reduction is acceptable, but not as good as the LMS algorithm, the Kalman filter or the WFLC algorithm. The noise reduction results with the LMS algorithm, the Kalman filter and the WFLC algorithm were similar to each other. Compared with each other, as concluded in Paper C, the implementation of the WFLC algorithm is moderate and the WFLC algorithm has better performance than the LMS algorithm and the Kalman filter on the reduction of physiological tremor and electrical noise. The KWT algorithm reduced the noise in the static state without interrupting the continuous movement signals.

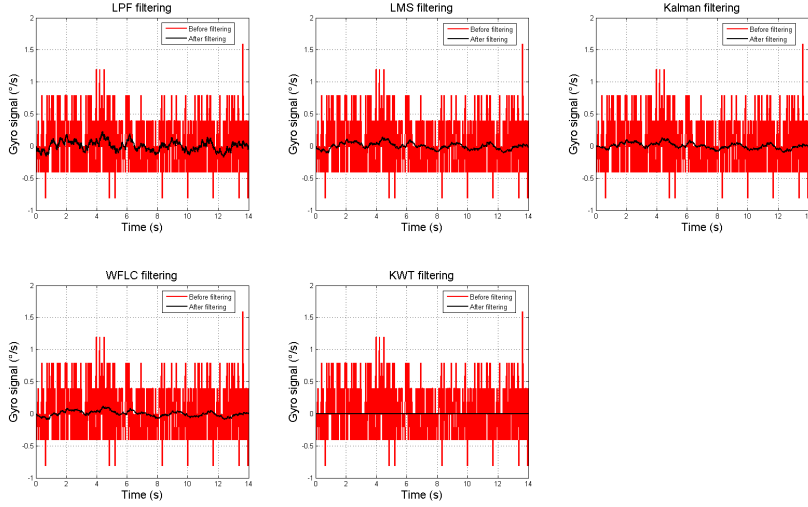


Figure 5.5: Noise reduction without movement by low-pass filter, LMS algorithm, Kalman filter, WFLC algorithm and KWT algorithm, respectively.

5.1.3 Summary

The performance summary of the algorithms for signal error reductions is shown in Table 5.1. High-pass filter, LMS algorithm, Kalman filter reduced drift/offset (as shown in Paper A). Low-pass filter, LMS algorithm, Kalman filter and WFLC algorithm reduced noise (as shown in Paper C). The KWT algorithm reduced the noise, offset and drift simultaneously (as shown in Paper D). These algorithms were developed with limited calculation capacity in real time. Due to the limitation of the small embedded sensor systems, some signal processing algorithms such as wavelet transform and Wiener filter were not included in this study. Although some signal processing algorithms, e.g. an improved wavelet threshold denoising method [39] and Extended Kalman filter [38], were developed for the MEMS sensor signal improvement, they require more calculations than the algorithms introduced in this thesis. The performances of the implemented signal processing algorithms were evaluated with drift/offset reduction, noise reduction, the implementation, delay and root-mean-square error (RMSE). The performance for the drift/offset and noise reduction, i.e. 'ok', 'good' or 'best', is evaluated based on the RMSE, the delay

time and the test feedback from the users. The RMSE were calculated from the signals without movements. The implementation is classified as easy, middle and complex, where ‘easy’ means the code was written in dsPIC in less than 20 rows based on mathematical algorithms, ‘middle’ means the code ranges from 20 to 40 rows and ‘complex’ means the code in more than 40 rows. The delay is regarded as the filter processing delay due to the calculation time that the algorithms need. It is a key factor to evaluate the algorithms in real time. User-perceived delay should be avoided by the systems at hand. Compared with the USB polling rate from 100 Hz to 1000 Hz and the corresponding maximal delay of 10 ms and minimal delay of 1 ms [59], and the user-unnoticeable delay of 150 ms on human-computer interactions [60], the maximal delay of 2 ms is acceptable by both the system and the users. According to the summary table, the Kalman filter was regarded as the best algorithm for drift reduction and WFLC for noise reduction. The KWT algorithm took advantage of Kalman and WFLC algorithm. With TWD algorithm, it achieved 100% noise reduction during the static state.

Table 5.1: Performance summary of the algorithms for signal error reduction.

Parameters	HPF	LPF	LMS	KF	WFLC	KWT
Drift/offset reduction	Ok	-	Good	Best	-	Best
RMSE	4.291	-	3.90	3.81	-	0
Noise reduction (movement)	-	Ok	Good	Good	Best	Best
Noise reduction (static state)	-	Ok	Good	Good	Good	Best
RMSE	-	0.075	0.049	0.044	0.042	0
Implementation	Easy	Easy	Middle	Complex	Complex	Complex
Delay (3-axes)	0.2 ms	0.2 ms	0.5 ms	1.0 ms	0.8 ms	2.0 ms

5.2 Improvement of position accuracy

This section addressed the second research question (Q2) on how to improve the position accuracy by MEMS accelerometer and MEMS gyroscope. There

are two main methods introduced to improve the position accuracy. One is the composition of low-pass filter, TWD method and velocity trend estimation method for MEMS accelerometer (Paper B). The other is the KWT algorithm for MEMS gyroscope (Paper D). Both of the two methods can reduce the accumulated error from digital integration.

5.2.1 MEMS accelerometer

The composition of low-pass filter, TWD method and velocity trend estimation method was designed to reduce the integration error during the position measurement with MEMS accelerometer. As shown in Figure 5.6, the green curve is the measured acceleration signal after unit conversion and offset reduction. The measurement was done with several up-and-down movements by a Test Rig for MEMS accelerometers [61]. The MEMS accelerometer was moved by Test Rig from one fixed point to 1 meter up to the point and then back to the fixed point. The movement was repeated 9 times. The red curve is calculated position without filtering and the black one is the result with filtering. The algorithm reduced the integration drift of MEMS accelerometer in one minute by about 20 meters. About 99.8% error reduction was achieved.

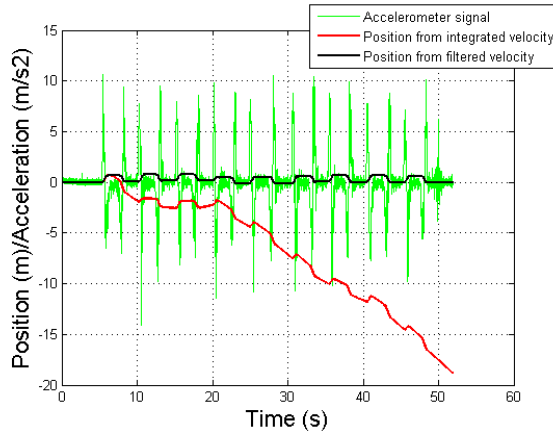


Figure 5.6: Improvement of position accuracy by low-pass filter, TWD method and velocity trend estimation method. The green curve is measured MEMS accelerometer signal [m/s^2]; the red one is position [m] without filtering; the black one is position [m] with filtering.

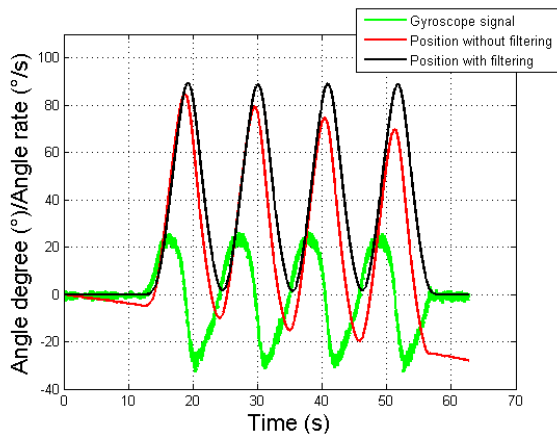


Figure 5.7: Angle calculated from MEMS gyroscope signals without drift. The green curve is measured MEMS gyroscope signal [$^{\circ}/s$]; the red one is angle [$^{\circ}$] without filtering; the black one is angle [$^{\circ}$] with filtering.

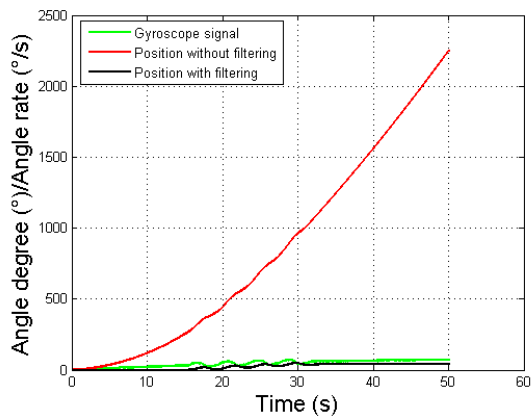


Figure 5.8: Angle calculated from MEMS gyroscope signals with drift. The green curve is measured MEMS gyroscope signal [$^{\circ}/s$]; the red one is angle [$^{\circ}$] without filtering; the black one is angle [$^{\circ}$] with filtering.

5.2.2 MEMS gyroscope

The KWT algorithm was developed to improve the position accuracy by MEMS gyroscope. As shown in Figure 5.7 and Figure 5.8, the green curve is the measured MEMS gyroscope signal. The measurement was done by pre-defined movements of Test Rig for MEMS gyroscope [33] in a slow mode (up to $15^\circ/\text{s}$). The test was performed from a fixed point to 90° from the point and then back to the start point. Totally four continuous rounds were performed from a stationary state and back to a stationary state.

As shown in Figure 5.7, the KWT algorithm reduced the integration drift of MEMS gyroscope signal without drift in one minute by about 25 degrees. 98.2% errors were reduced.

As shown in Figure 5.8, the KWT algorithm reduced the integration drift of MEMS gyroscope signal with drift in one minute by about 2500 degrees. 99.9% errors were reduced.

Chapter 6

Conclusion and future work

6.1 Conclusion

Signal processing is regarded as an important and necessary way to improve the performances of MEMS sensor based human motion analysis system by reducing the MEMS sensor signal errors such as noise, offset and drift, especially temperature drift and integration drift. Due to MEMS sensor characteristics and the measurement environment in human motion analysis, the signal processing algorithms with MEMS gyroscopes and MEMS accelerometers for noise reduction, offset/drift estimation and reduction, position accuracy and system stability were presented in this licentiate thesis. The signal processing algorithms in this study fulfilled the requirements in the following aspects:

- Be able to minimize signal errors and improve the signal quality.
- Be simple enough for implementation and comply with the system requirements.
- Be compatible for different MEMS sensors.

High-pass filter, LMS algorithm and Kalman filter were found useful to reduce the offset and drift, especially the temperature drift in a MEMS gyroscope based system. On the other hand, low-pass filter, LMS algorithm, Kalman filter and WFLC algorithm were used for noise reduction. Comparison study showed that the most effective method to reduce temperature drift and noise including human physiological tremor and electrical noise were Kalman filter and WFLC algorithm respectively. The methods like threshold with delay

method and velocity estimation method were designed to reduce the accumulated calculation error during digital integration process. Together with the low-pass filter and Trapezoidal method, the two methods improved the result of position measurement with MEMS accelerometer greatly. The reduction of the integration drift was about 20 meters in one minute. The threshold with delay method was developed based on the normal threshold method, but with the improvement on non-interrupt of the continuous movement signals. The KWT algorithm reduced noise, offset and drift together and improved the static stability and position accuracy of a MEMS gyroscope based system substantially. 100% error reduction during the static state, 98.2% position error correction in the case of movements without drift, and 99% with drift were achieved. The hardware requirement of limited calculation capacity in real time was achieved with maximal computational time of 2 ms, which was acceptable by both hardware system and users.

6.2 Future work

High-pass filter, low-pass filter, LMS algorithm, Kalman filter, WFLC algorithm have been demonstrated their feasibility to reduce signal errors in the MEMS gyroscope based system. They promoted the signal improvement of MEMS sensor based motion analysis systems. More algorithms like Cascaded Integrator-Comb (CIC) filter, Ensemble Empirical Mode Decomposition (EEMD) etc. are foreseen to be developed, implemented, tested and evaluated with these existing algorithms.

The threshold with delay method and the velocity estimation method were used to reduce the integrated drift during the position measurement with MEMS accelerometer. The KWT algorithm improved the angular position accuracy during the Test Rig movement with MEMS gyroscope. The future work includes the development of algorithms for movement measurement and position calculation with MEMS sensors. The hardware system with MEMS sensors should be developed as small and light as possible, and the position accuracy is urgently needed to be improved.

Furthermore, since there are initialized parameters in many algorithms and the selection of the parameters has an impact on the performance of the algorithms, optimal parameter selection of these developed algorithms is another important issue to be addressed in the future.

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